

Gradient Boosting - Algorithm (Classification)

This is **Sequentially stage-wise Modeling**

weak learners

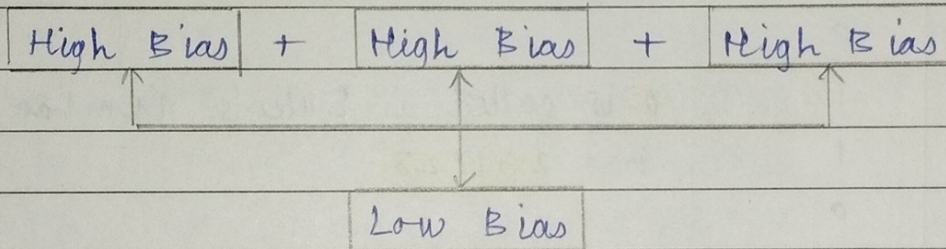
Base Model $f_0(x)$	Decision Tree 1 $f_1(x)$	Decision Tree 2 $f_2(x)$	Decision Tree 3 $f_3(x)$
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$$F(x) = f_0(x) + \alpha \cdot f_1(x) + \alpha \cdot f_2(x) + \alpha \cdot f_3(x)$$

This is called as **Additive Modeling**

(A combination of multiple weak learners)

weak learners have **high bias** :- so with the help of **additive modeling** we **combine multiple weak learners** of **High bias** to make it **Low Bias**



Small Classification Dataset

cgpa	iq	is_placed
8.82	118	0
6.38	125	1
5.39	99	1
5.50	106	1
6.39	148	0
9.13	148	1
7.17	147	1
7.72	72	0

BM + DT1 + DT2

In Regression
this model
is **mean**
of dependent
feature

In Classification
we calculate
log(odds)

$$\log(\text{odds}) \rightarrow \log\left(\frac{\#1}{\#0}\right)$$

#1 :- No of 1

#0 :- No of 0

Stage 1: Calculate $\log(\text{odds})$ and probability of $\log(\text{odds})$

* $f_0(x) = \log\left(\frac{n_+}{n_-}\right)$ $n_+ = \text{number of positive class}$
 $n_- = \text{number of negative class}$

Sample Datasets

cgpa	iq	is-placed
6.82	118	0
6.36	125	1
5.39	99	1
5.50	106	1
6.39	148	0
9.13	148	1
7.17	147	1
7.72	72	0

$$\log\left(\frac{5}{3}\right) = 0.510826$$

∴ We use Natural logarithm ($\ln$)
 - which is $\log$ base e

e is called as Euler's number which
 $e = 2.71828$

— this is f_0 (initial)

cgpa	iq	is-placed	$p_{\text{pred}}(\log\text{-odds})$
6.82	118	0	0.510826
6.36	125	1	0.510826
5.39	99	1	0.510826
5.50	106	1	0.510826
6.39	148	0	0.510826
9.13	148	1	0.510826
7.17	147	1	0.510826
7.72	72	0	0.510826

$p_{\text{prediction}}$



after this we
 usually calculate
 residuals in regression



but in classification
 we take probability
 of initial prediction
 or log-odds

The is-placed ← because ←
 which is our Dependent feature
 gives us kind of probability

find

(This is the reason behind why we cannot, residuals
 for classification) because probability and log-odds
 cannot be used to find residuals or even combine them

★ Now How we calculate probability using this formula :- $p = \frac{1}{1 + e^{-\log(\text{odds})}}$

$$p = \frac{1}{1 + e^{-0.510826}} = 0.625$$

(y) actual value

(p) predicted probability

cgpa	ia	is-placed	pred1 (log-odds)	pred1 (probability)
6.82	118	0	0.510826	0.625
6.36	125	1	0.510826	0.625
5.39	99	1	0.510826	0.625
5.50	108	1	0.510826	0.625
6.39	148	0	0.510826	0.625
9.13	148	1	0.510826	0.625
7.17	147	1	0.510826	0.625
7.72	72	0	0.510826	0.625

regression → Mean

classification → log(odds)

★ Now we calculate residual error

$$\text{Residual} = y - p$$

Where : y = actual label or value (0 or 1)

p = predicted probability

cgpa	ia	is-placed	pred1 (log-odds)	pred1 (prob)	res1
6.82	118	0	0.510826	0.625	-0.625
6.36	125	1	0.510826	0.625	0.375
5.39	99	1	0.510826	0.625	0.375
5.50	108	1	0.510826	0.625	0.375
6.39	148	0	0.510826	0.625	-0.625
9.13	148	1	0.510826	0.625	0.375
7.17	147	1	0.510826	0.625	0.375
7.72	72	0	0.510826	0.625	-0.625

Stage 2: Train Model 2 (Decision Tree) $F_1(x)$:- where
 input = Independent feature
 output = residual errors (not Dependent feature)

- In Regression the output of Decision Trees was the leaf's values itself
- In ~~Classification~~ the output of Decision Tree leafs are probabilities → since the values are probabilities we cannot add them to the log(Odds)
- In Regression we use this formula to find the optimal leaf value to minimize error!

$$\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, F_{m-1}(x_i) + \gamma)$$

→ In ~~Classification~~

$\sum \text{Residual}$

$$\sum [P_{\text{previous Prob}} \times (1 - P_{\text{previous Prob}})]$$

⇒ This formula gives you the optimal log-odds value that each leaf should output to minimize the loss!

Steps: How to use the formula to find log-odds value

→ For each leaf, identify which instances belong to it, then:

Step 1: calculate Residual for each instance

⇒ Residual = $y - p$

Step 2: calculate $p \times (1-p)$ for each instances
 $p \times (1-p) = \text{Previous Probability} \times (1 - \text{Previous Probability})$

Step 3: Sum them up

- Numerator = Sum of all residuals in that leaf
- Denominator = Sum of all $p \times (1-p)$ in that leaf

Step 4: Divide

$$r = \text{Numerator} / \text{Denominator}$$

Eg Dataset and Tree (How log Odds) value is found)

cgpa	iq	is_placed	pred(log-odds)	pred(p prob)	res
6.82	118	0	0.510826	0.625	-0.625
6.36	125	1	0.510826	0.625	0.375
5.39	99	1	0.510826	0.625	0.375
5.50	106	1	0.510826	0.625	0.375
6.39	148	0	0.510826	0.625	-0.625
9.13	148	1	0.510826	0.625	0.375
7.17	147	1	0.510826	0.625	0.375
7.72	72	0	0.510826	0.625	-0.625

⇒ When we give :-

X :- Independent features

y :- Residuals

node # 0

cgpa ≤ 6.35

squared_error = 0.234

samples = 8

Value = 0.0

⇒ We got something like

this and this uses

the formula to find

log-odds for every

instance

node # 1

squared_error = 0.0

samples = 3

Value = 0.375

node # 2

iq ≤ 132.5

squared_error = 0.24

samples = 5

Value = -0.225

⇒ When we give Model 2 (Decision Tree) our X and r it does best split with the help of MSE (Mean Squared Error) based on: Input as X (Independent feature) and r (Residuals)

⇒ After splitting it automatically identifies which best belongs to which instance
(It can track which instances go to which best automatically)

⇒ ~~After split~~ Once it decides which instances it uses this formula:-

$$\frac{\sum \text{Residual}}{\sum [P_{\text{previous prob}} \times (1 - P_{\text{previous prob}})]}$$

⇒ This formula gives optimal log(odds) value

★ Calculation for node 1 (3 samples)

→ Node #1 has three samples

Actual (y)	pnev(log-odds)	pnev(prob)	residual
1	0.510826	0.625	0.375
1	0.510826	0.625	0.375
1	0.510826	0.625	0.375

step 1: Calculate Numerator (Sum of Residuals)

$$\text{Numerator} = 0.375 + 0.375 + 0.375 = 1.125$$

step 2: Calculate Denominator (sum of $p \times (1-p)$)

$$p = 0.625$$

$$(1-p) = 1 - 0.625 = 0.375$$

$$p \times (1-p) = 0.625 \times 0.375 = 0.234375$$

$$\text{Denominator} = 0.234375 + 0.234375 + 0.234375 = 0.703125$$

Step 3 : Calculate γ

$$\gamma = \text{Numerator} / \text{Denominator}$$

$$\gamma = 1.125 / 0.703125$$

$$\gamma = 1.6$$

What this means :

↳ This $1.6 \log(\text{odds})$ adjustment will make the predictions for these 3 instances more accurate (closer to their actual value of 1)

⇒ Unlike in Regression we directly use leaf's output value but in classification we need to add it up with previous $\log(\text{odds})$

learning rate ←

$$\Rightarrow \text{Model 2} = \text{Model 1}(\log\text{-odds}) + \text{Tree output} \cdot \text{Tree output}$$

$$= 0.510826 + 0.1 \cdot 1.6$$

$$= 0.66$$

↳ This is Model 2 $\log(\text{odds})$ value
(next-page value) ←

cgpa	iq	u-placed	pred1(log-odds)	pred1(prob)	res1	pred2(log-odds)
6.82	118	0	0.510826	0.625	-0.625	0.245
6.36	125	1	0.510826	0.625	0.325	0.445
5.39	99	1	0.510826	0.625	0.375	0.445
5.50	106	1	0.510826	0.625	0.375	0.445
6.39	148	0	0.510826	0.625	-0.625	0.245
9.13	148	1	0.510826	0.625	0.375	0.445
7.17	147	1	0.510826	0.625	0.375	0.445
7.72	72	0	0.510826	0.625	-0.625	0.245

⇒ To prevent ~~one~~ from overfitting we can add a learning rate when calculating $\log(\text{odds})$ ~~or~~ when ~~calculating residuals~~.

pred2 (log-odds)	pred2 (probability)
0.244	0.5607
0.67	0.6615
0.67	0.6615
0.67	0.6615
0.528	0.6290
0.528	0.6290
0.528	0.6290
0.244	0.5607

⇒ After getting pred2 (log-odds) value we gonna get its **probability** using this **formula** $\frac{1}{1 + e^{-\log(\text{odds})}}$

⇒ After finding probability we gonna calculate its residual error for training the next model

residuals 2	residual 1
-0.5607	-0.625
0.3385	0.375
0.3385	0.375
0.3385	0.375
-0.629	-0.625
0.371	0.375
0.371	0.375
-0.5607	-0.625

After comparing residual 1 & 2 we came to know that by **adding multiple estimations** we can reduce this value and bring it to **zero** or **very near to zero**

residual = $y - p$

that means our predicted value is connect output value

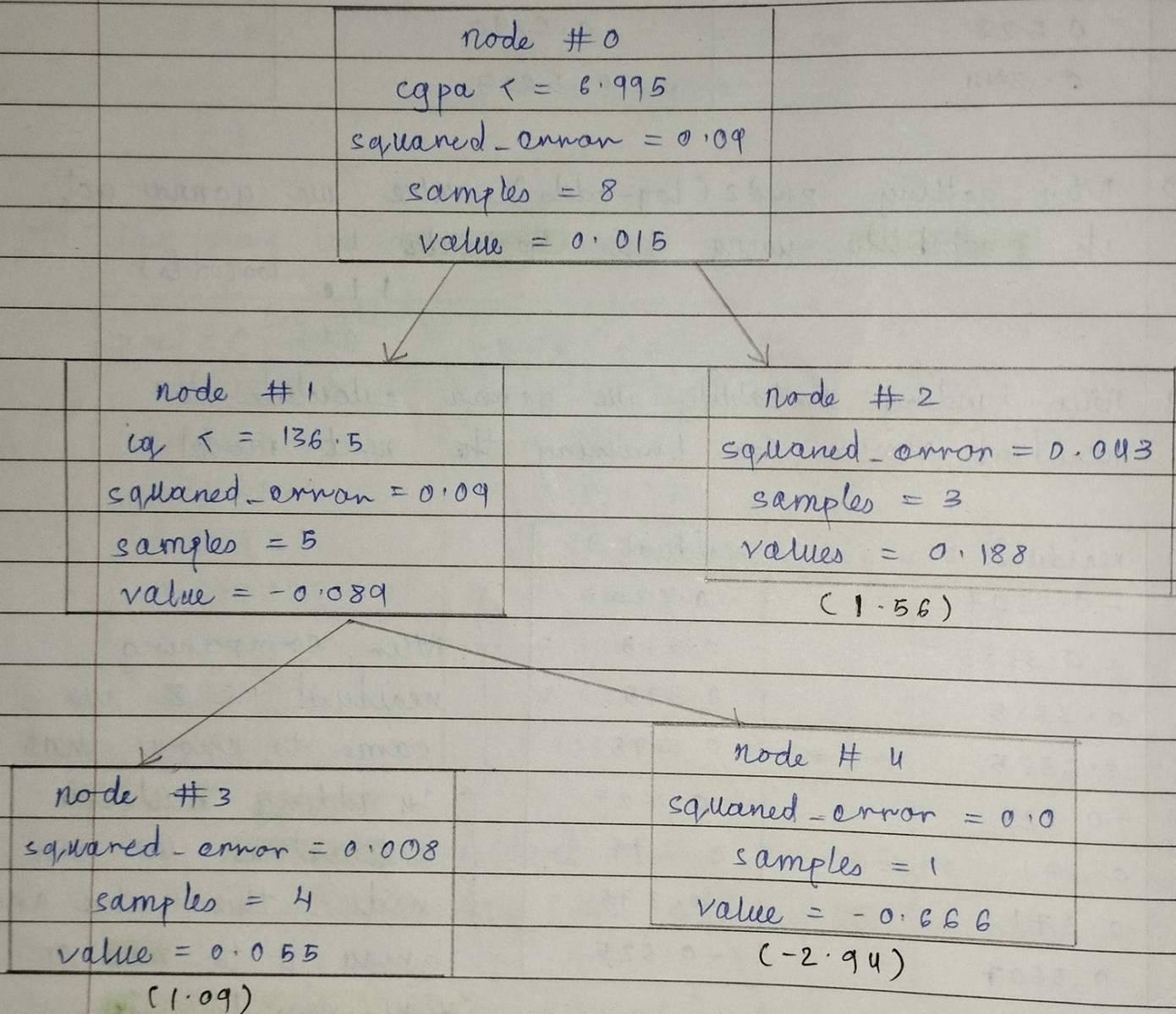
Formula :- $\hat{y}_m(x) = \hat{y}_0 + \sum_{i=1}^m v_i \cdot h_m(x)$

Stage 3:- Construct a 3rd Model (2nd Decision Tree)

X = Independent Features

r = residual 2

⇒ This is how the decision tree looks like when input is given as Independent Features and output as residuals



⇒ Here, the model gives output in log(odds)

⇒ And, again we need to convert this into probability to in order to prediction or calculate residuals

⇒ This loop continues until it ~~reaches~~ reaches the last model (estimator)

⇒ Now to calculate this node's output we use this formula :-

$$\frac{\sum \text{Residual}}{\sum [\text{Previous Prob} \times (1 - \text{Previous Prob})]}$$

⇒ By using this formula, we get outputs of every node

node 2 = 1.56

node 3 = 1.09

node 4 = -2.94

Note :- When splitting it automatically tracks which node (leaf) value belongs to which instances

→ 1st instance → belongs to 3rd Node this is

Model 2 (log-odds) :- 0.244 logCdds) value

$$0.244 + 0.1 \times (1.09) = 0.353$$

↪ We gonna add them-up :- ~~0.244 + 0.1 \times (1.09)~~ = 0.353

→ After finding log-odds we use this formula to convert them into probability

→ formula :-
$$p = \frac{1}{1 + e^{-(\text{log-odds})}}$$
 = 0.587 this is Probability value

Note :- 0.244 is already calculated using model 1 + 0.1 x Model 2 we just need to do with the current one not allow again

Final Output :- Formula

↪
$$F(x) = f_0(x) + c \cdot f_1(x) + d \cdot f_2(x) + \dots + e \cdot f_m(x)$$

↪ After this we need to calculate its probability :-

$$p = \frac{1}{1 + e^{-\text{logCdds}}}$$